

IV.A. Industry Computerization and Task Trends over Four Decades

We begin by estimating a model for the within-industry relationship between computer adoption and task change over four decades. Specifically, we fit the equation

$$(13) \quad \Delta T_{jk\tau} = \alpha + \phi \Delta C_j + \varepsilon_{jk\tau},$$

where $\Delta T_{jk\tau} = T_{jk\tau} - T_{jkt}$ is the change in industry j 's input of task k between years t and τ and ΔC_j is the annual change in the percentage of industry workers using a computer at their jobs between 1984 and 1997 (estimated from the October Current Population Survey supplements). While our model predicts that computer adoption should be highly correlated with industry task change during the computer era, we would not anticipate a similar relationship for the precomputer era of the 1960s. Accordingly, we estimate equation (13) separately for each of the four decades in our sample. This allows us to confirm that the relationship between computerization and industry task change in the 1970s–1990s does not reflect trends that predate the computer era.¹⁸

Estimates of equation (13) for each of the four task measures are given in Table III. The first two panels show that during the 1970s, 1980s, and 1990s, rapidly computerizing industries raised their input of nonroutine analytic and interactive tasks significantly more than others. For example, the point estimate of 12.04 (standard error of 4.74) in column 1 of panel A indicates that between 1990 and 1998, a 10 percentage point increase in computer use was associated with a 1.2 centile annualized increase in labor input of nonroutine analytic tasks. To evaluate the magnitude of this relationship, note that the mean annualized industry level rise in nonroutine analytic task input during 1990–1998, tabulated immediately below the regression estimate, was 2.5 percentiles. By comparison, the intercept of the bivariate regression for this period is 0.1. Hence, almost the entirety of the observed within-industry change in nonroutine analytic input is “explained” by the computerization measure. Similar comparisons confirm that the relationship between industry computerization and rising input of nonroutine interactive and analytic

18. As noted by Autor, Katz, and Krueger [1998] and Bresnahan [1999], the era of rapid computer investment began in the 1970s. Desktop computing became widespread in the 1980s and 1990s.

TABLE III
COMPUTERIZATION AND INDUSTRY TASK INPUT, 1960–1998
DEPENDENT VARIABLE: $10 \times$ ANNUAL WITHIN-INDUSTRY CHANGE IN TASK INPUT,
MEASURED IN PERCENTILES OF 1960 TASK DISTRIBUTION

		1. 1990– 1998	2. 1980– 1990	3. 1970– 1980	4. 1960– 1970
A. Δ Nonroutine analytic	Δ Computer use	12.04	14.02	9.11	7.49
	1984–1997	(4.74)	(4.97)	(4.17)	(5.28)
	Intercept	0.07	–0.66	–0.26	–0.55
		(1.00)	(1.03)	(0.86)	(1.05)
	R^2	0.04	0.05	0.03	0.01
B. Δ Nonroutine interactive	Weighted mean Δ	2.45	2.05	1.48	0.83
	Δ Computer use	14.78	17.21	10.81	7.55
	1984–1997	(5.48)	(6.32)	(5.71)	(6.64)
	Intercept	1.02	1.46	2.35	0.10
		(1.15)	(1.31)	(1.17)	(1.32)
C. Δ Routine cognitive	R^2	0.05	0.05	0.03	0.01
	Weighted mean Δ	3.94	4.79	4.42	1.49
	Δ Computer use	–17.57	–13.94	–11.00	–3.90
	1984–1997	(5.54)	(5.72)	(5.40)	(4.48)
	Intercept	–0.11	0.63	1.63	1.78
D. Δ Routine manual		(1.17)	(1.19)	(1.11)	(0.89)
	R^2	0.07	0.04	0.03	0.01
	Weighted mean Δ	–3.57	–2.07	–0.47	1.06
	Δ Computer use	–24.72	–5.94	–6.56	4.15
	1984–1997	(5.77)	(5.64)	(4.84)	(3.50)
	Intercept	1.38	–0.16	2.09	0.85
		(1.22)	(1.17)	(0.99)	(0.70)
	R^2	0.12	0.01	0.01	0.01
	Weighted mean Δ	–3.50	–1.31	0.84	1.62

n is 140 consistent CIC industries. Standard errors are in parentheses. Each column of panels A–D presents a separate OLS regression of ten times the annual change in industry-level task input between the endpoints of the indicated time interval (measured in centiles of the 1960 task distribution) on the annual percentage point change in industry computer use during 1984–1997 (mean 0.193) and a constant. Computer use is the fraction of industry workers using a computer at their jobs, estimated from the October 1984 and 1997 CPS samples. Estimates are weighted by mean industry share of total employment in FTEs over the endpoints of the years used to form the dependent variable. Samples used are Census 1960, 1970, and 1980 and CPS MORG 1980, 1990, and 1998. See Table I and Appendix 1 for definitions and examples of task variables.

tasks is economically large and statistically significant at conventional levels in each of the three most recent decades.

Panels C and D of the table provide analogous estimates for the two routine task measures. As predicted by the conceptual model, the relationships between industry computerization and changes in routine task input are uniformly negative in the 1970s, 1980s, and 1990s. These relationships are also economi-

cally large and in most cases statistically significant. For example, the computerization measure explains the entirety of the within-industry decline in routine task input during the 1990s, and more than explains this decline in the 1980s and 1970s.

A notable pattern for all four task measures is that the relationship between computerization and industry task change tends to become larger in absolute magnitude with each passing decade. This suggests a secularly rising relationship between computerization and task change. The final column of Table III tests for this rise by estimating equation (13) for the 1960s, a decade during which computerization is unlikely to have strongly influenced task demands. Reassuringly, there are no significant relationships between computerization and task change in this decade. And in one case, the coefficient is of the opposite sign as in later decades. Hence, these estimates suggest that the relationship between industry task shifts and computer adoption either commenced or substantially accelerated during the computer era, and not before.¹⁹

IV.B. Using Composite DOT Variables

Though we view the selected task measures as the most appropriate available from the DOT, we are sensitive to the concern that the choice of variables could be viewed as arbitrary. One way to test their appropriateness is to use alternative composite variables. We used principal components analyses (PCA) to pool variation from each selected DOT task measure with several other plausible alternatives and estimated equation (13) using these composites.²⁰ The details of our compositing exercise are provided in the Data Appendix, and the results of the composite estimation are found in Appendix 2. A limitation of this exercise is that the variables used in the composites do not in our view

19. We also estimated the models in Table III separately by gender and for manufacturing and nonmanufacturing sectors. The pattern of results is similar in all cases. For both genders, computer investment is a significant predictor of reductions in routine labor input of cognitive and manual tasks and increases in nonroutine analytic input. For females the relationship between computerization and nonroutine interactive tasks is positive but insignificant. The magnitude of the relationship between computerization and nonroutine tasks is somewhat larger in manufacturing than nonmanufacturing, and the reverse is true for routine tasks. Further details are available from the authors.

20. The PCA extracts eigenvectors that maximize common variation among selected measures, each of which is standardized with mean zero and variance one, subject to the constraint that the sum of squared weights in the eigenvector equals one. It can be shown that if measurement error in the selected variables is classical (i.e., white noise), the PCA extracts maximal nonerror variation.

correspond as closely to the intended construct as our primary measures.

As visible in the table, the qualitative trends in the composite relationships are comparable to those using our preferred measures in Table III. In particular, industry computerization is associated with sharp declines in routine cognitive and manual labor inputs and growth in nonroutine analytic and interactive labor inputs. Moreover, these relationships typically become stronger in successive decades. Contrary to expectations, however, the composite measure for routine cognitive input is only significant in the most recent decade and the composite measure for nonroutine interactive input is statistically significant in the 1960s. Thus, while our results are generally robust to variable choice, this exercise underscores that variable choice does matter. A data source specifically designed to measure changes in workplace input of routine and nonroutine cognitive tasks over a long time horizon would clearly provide a more complete test of the model. Given the absence of such a data source for the United States, we view the evidence provided by the DOT as uniquely informative.²¹

IV.C. Employing Contemporaneous Measures of Computer and Capital Investment

A limitation of the CPS computer measure used so far is that it is only available for the 1980s and 1990s. To provide more comprehensive measures of computer and capital investment available for the entire 1959–1998 period, we draw on the National Income and Product Accounts (NIPA), which provides detailed data on capital stocks across 42 major industries excluding government [U. S. Department of Commerce 2002a, 2002b]. As a measure of industry computerization, we calculated the log of real investment in computer hardware, software and peripherals per full-time equivalent employee (FTE) over the course of each decade. To distinguish the relationship between task change and computerization from overall capital-skill complementarity [Griliches 1969], we construct two variables to control for capital

21. Spitz [2003] studies the predictions of our task model using German data from 1979–1999, which contains far more detailed and precise information on workplace tasks than is available from the DOT. Consistent with the predictions of the model, Spitz reports that computer capital substitutes for repetitive manual and repetitive cognitive skills and complements analytical and interactive skills. See also Bartel, Ichniowski, and Shaw [2000] and Ichniowski and Shaw [2003] for quantitative and case study evidence.